Mathapathi Sai Manjunath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.73
SRI CHAITANYA JUNIOR COLLEGE	Class XII (BIEAP)	2023	95.90%
SRI CHAITANYA EM HIGH SCHOOL	Class X (SSC)	2021	10.00%

EXPERIENCE

Tower Research Capital, *Core Engineering Intern*

Project-1: *Close Price Feeder* | language: ***Rust***

- Developed a **Rust** application to consume **latest/daily close prices** from upstream price feeds and persist them in a database
- Implemented and deployed the feeder for **equity** markets across APAC, EMEA and USA covering **8 markets**
- Achieving **87% coverage** of the equities available at TRC, capturing **close prices** for approx. **42k+ equities**
- Extending Close Price Feeder support to **Crypto markets**

Project-2: *Simnetizing P&L Server and Support tools | language: **C++***

- Ported P&L Server to TRC internal Simnet **testing framework** which enables **deterministic** and **reproducible** testing
- **Simnetized** its **admin control** tool, **client tool** to enable **end-to-end testing**

PROJECTS

Multi-Client Quiz System with LLM integration

Course Project under Prof. Kotaro Kataoka and Prof. Praveen Tammana / Github

- Developed a **TCP** based **client-server** quiz system in **C++** with **LLM integration** for question generation
- Designed a **custom** application layer **protocol** with **keep-alive**/alive-ok messaging for **session management**
- Developed **caching** on the server side to **reduce LLM traffic** for popular quiz genre question generation
- Analyzed **network traffic** and **system performance** using Wireshark, measuring **throughput** and **request-response latency** under varying client **concurrency**

Network Fairness on Linux using eBPF

Course Project under Prof. Kotaro Kataoka and Prof. Praveen Tammana | Github

- Developed an **eBPF-based network traffic control system** for enforcing per-cgroup bandwidth policies on Linux
- Implemented **cgroup-aware packet filtering and rate limiting** using eBPF, by maintaining per-cgroup traffic-control state
- Built a **C-based userspace controller** to configure rate-limiting parameters and evaluated traffic enforcement

Memory Management System Calls in GemOS

Course Project under prof. Saurabh kumar / Github

- Implemented system calls (**mmap, munmap, mprotect**) to efficiently manage virtual memory of a process
- Implemented **vm_area_pagefault** system call to allocate physical frames for a process using **lazy allocation**
- Extended munmap and mprotect to **update page table enteries** on unmapping and protection changes

RISC-V Assembly Simulator with Cache

Course Project under Prof. Rajesh Kedia / Github

- Developed an **RV64I** simulator in **C++** with support for step by step execution and **register state tracking**
- Integrated **breakpoint handling** with support for break/continue execution at **user defined** breakpoints

- Developed a **configurable cache** simulator for memory instructions with **cache statistics** support
- Supported multiple **replacement policies** (LRU, LFU, Random), **associativity levels**, and **cache sizes**

Infinite Precision Arithmetic

Course Project under Prof. Ramakrishna Upadrasta and Prof. Rajesh kedia / Github

- Designed and implemented a C++ library for **arbitrary precision arithmetic** on integers and real numbers
- Designed **custom classes** to support operations (add, subtract, multiply, divide) on **unbounded** numeric types
- Applied **object oriented design** and encapsulated classes within **namespaces** for modularity and reuse

SKILLS

Programming Languages: C, C++, Rust, RISC-V Assembly, Bash Scripting, MySQL

Tools: Git, LaTeX

Competitive Programming: Codeforces - Max Rating (1220), Leetcode (140+ solved)

RELEVANT COURSES

Core Courses: Computer Architecture, Data Structures, Operating Systems, Database Management Systems, Compilers, Computer Networks (A), Foundations of Machine Learning, Deep Learning, Systems for Machine Learning

Other Relevant Courses: Probability & Random Variables, Linear Algebra, Statistical Learning Theory, Convex Optimization I & II

LEADERSHIP & ACADEMIC ROLES

Class Representative for CSE'27 Batch

Teaching Assistant for Introduction to Programming (ID 1063) Course

Teaching Assistant for Introduction to Software Programming (HT5023) course

SCHOLASTIC ACHIEVEMENTS

Qualified for **NTSE level 2** with 165 Marks (70 - SAT, 95 - MAT)

EXTRACURRICULAR

Student Volunteer in National Service Scheme (NSS)